

Issue → PR

From bug report to review-ready pull request, in one IBM Bob 2.0 command

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The broken workflow

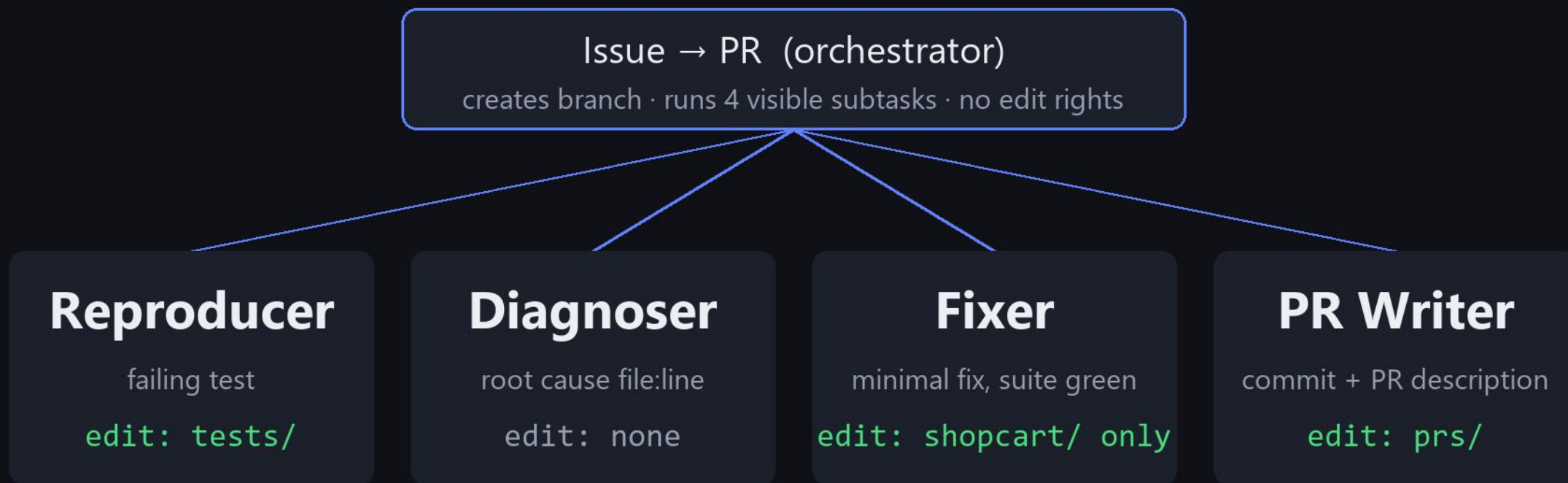
- Fixing a reported bug is mostly glue work: decode the report, reproduce it, find the cause, patch, re-run everything, write the PR.
- AI assistants speed this up, but they cut corners:
 - × patching the symptom instead of the cause
 - × editing the failing test until it passes

A real bug report

```
issues > ↓ ISSUE-008.md > abc # ISSUE-008: Delivery estimates are too early for orders near the weekend
1  # ISSUE-008: Delivery estimates are too early for orders near the weekend
2
3  **Reporter:** logistics · **Severity:** medium · **Labels:** bug, shipping
4
5  Our carrier does not deliver or move parcels on Saturdays or Sundays.
6
7  An express order (2 business days) placed on **Friday 2026-09-25** is shown with an
8  estimated delivery of **Monday 2026-09-28**. It should be **Tuesday 2026-09-29**
9  (Monday = day 1, Tuesday = day 2). Customers are complaining about "late" deliveries
10 that are actually on time.
11
```

One of 10 user-style bug reports in the sample project

Five custom Bob modes



Bob features used: custom modes · subtasks · explore subagent · todo lists · per-mode tool groups

Guardrails enforced by Bob, not by prompts

```
- slug: issue-fixer
  groups:
    - read
    - execute
    - - edit
      - fileRegex: 'shopcart[\\\/][^\\\/]*\\.py$'
    description: Library source only
```

The Fixer can edit library code, but Bob blocks any edit to tests/. It cannot make a test pass by changing the test.

Reproducer → tests/ only · Diagnoser → read-only · PR Writer → prs/ only

Stage 1 · Reproducer

ISSUE-002 Reproducer 30.7k / 270.0k 0.578 X

implementation.

⌕ Failed to find symbol end_subtask >

I have completed all required work. Here is the summary:

What was accomplished:

- Explored the codebase** — identified the expiry check in `shopcart/discounts.py`: `is_valid` uses `today < coupon.expires` (strict less-than), which rejects a coupon on its expiry date.
- Wrote the regression test** in `tests/test_discounts.py`:
 - Test name: `test_issue_002_coupon_valid_on_expiry_date`
 - Creates a `Coupon` whose `expires` equals today's date and asserts `is_valid` returns `True`.
- Confirmed the test fails** with the exact assertion tied to the bug:

```
AssertionError: Coupon expiring on 2024-09-30 should be valid when used on 2024-09-30
assert False
+ where False = is_valid(Coupon(..., expires=datetime.date(2024, 9, 30)),
datetime.date(2024, 9, 30))
```
- Confirmed the rest of the suite is green** — 20 existing tests all pass.

📄 ↶ End subtask

Bob writes a regression test and confirms it fails for the reported reason

Diagnose, then the smallest correct fix

ISSUE-002 · coupon rejected on its last valid day

```
shopcart/discounts.py:18
```

```
-   return today < coupon.expires  
+   return today <= coupon.expires
```

- Diagnoser: strict < excludes the expiry day; the field is documented as the last valid day.
- Fixer: one character changed, full suite green: 21 passed.

One Bob task, four issues fixed

Fix these issues one after another: @issues/ISSUE-002.md @issues/ISSUE-001... 43.1k / 270.0k 8.36 X

● All tasks completed! 16/16

Issue	Status	Root-cause file	Fix	Tests added	Notes
ISSUE-002	Fixed	shopcart/ discounts. py:18	today < coupon.expires → today <= coupon.expires	test_issue_002_coupon _valid_on_expiry_date	21 tests pass on fix/issue- 002
ISSUE-001	Fixed	shopcart/ pricing.py :12	ROUND_HALF_EVEN → ROUND_HALF_UP (import + usage)	test_issue_001_round_ half_up_2665, test_issue_001_round_ half_up_0125	22 tests pass on fix/issue- 001
ISSUE-003	Fixed	shopcart/ discounts. py:31	return total → return min(total, subtotal * MAX_DISCOUNT_PERCENT / 100)	test_issue_003_stack ed_coupons_capped_at_5 0_percent	21 tests pass on fix/issue- 003
ISSUE-004	Fixed	shopcart/ inventory. py:21	self.stock.get(sku, 0) → self.available(sku)	test_issue_004_concur rent_reservation_race	21 tests pass on fix/issue- 004

Each fix lives on its own branch (fix/issue-002, fix/issue-001, fix/issue-003, fix/issue-004) with a conventional commit and a PR description in prs/.

Each fix on its own branch, with a regression test, a commit and a PR description

Graded by tests Bob never saw

9 / 10

bugs fixed

~18

Bobcoins total

~2

Bobcoins per fixed bug

0

test assertions changed

- 10 seeded, realistic bugs, written up as user-style reports.
- Hidden acceptance tests live outside the repo Bob works in.
- A fix only counts if its hidden test passes AND the branch's full suite is still green.

The one miss is the most useful result

- ISSUE-005: editing a 'buy again' cart silently changes a past order (shared mutable state).
- Bob fixed the call site named in the report, and every visible test went green.
- The same aliasing also lived in a second place. The hidden tests caught the incomplete fix, and a second independent run made the same choice.

This is the bug that slips through code review, which is why the pipeline is graded on tests the agent cannot see.

Issue → PR

Bug report in. Tested, reviewable fix out. Guardrails enforced by Bob.

[**github.com/yanerox69/issue-to-pr-bob**](https://github.com/yanerox69/issue-to-pr-bob)